

Raja Satya Vardhana Reddy Thippareddy

AI/ML Developer | Python Developer | Applied AI Developer

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PROFESSIONAL SUMMARY

AI/ML and Python-focused developer building practical systems across healthcare AI simulation, multimodal assistants, big-data prediction, NLP, computer vision, IoT automation, and real business websites. Background combines B.Tech IT, Sheridan IoT with ML, and Humber AI with ML, with hands-on project evidence across code, reports, dashboards, videos, and live deployments.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, TypeScript, HTML, CSS

AI / ML: Machine learning, deep learning, NLP, computer vision, LLM evaluation, prompt engineering, transformer foundations

Libraries: pandas, NumPy, scikit-learn, TensorFlow/Keras, PyTorch, Hugging Face, OpenCV, DeepFace, MediaPipe, dlib

Data / Cloud: PySpark, HDFS, GCP VM, Elasticsearch, Kibana, Logstash, Jupyter, Git/GitHub, Vercel

Applied AI: OpenAI APIs, Whisper concepts, Google Vision/Gemini concepts, Ollama, OpenWebUI, local LLM deployment

IoT / Embedded: ESP32, ultrasonic sensors, servo control, Blynk IoT, sensor integration, room/visitor monitoring

EXPERIENCE

AI Evaluator / Performance Analyst (Freelance) | Outlier AI Dec 2025 - Present

- Evaluated and ranked LLM outputs for reasoning, instruction-following, factual accuracy, tone, and safety using structured rubrics.
- Identified hallucinations, inconsistent responses, and quality patterns to support model improvement workflows.
- Performed prompt evaluation, multimodal annotation, classification, and structured feedback tasks in guideline-driven AI workflows.

Crew Trainer | Tim Hortons, Ontario Jan 2024 - Aug 2025

- Promoted to trainer within six months; supported onboarding, customer operations, cash handling, inventory, and quality control.
- Built reliability, communication, and operational discipline in a high-volume 24/7 service environment.

Solo Web / Automation Builder | The Dansvilla Studio, Ottawa 2026

- Built and improved a live dance-studio website and inquiry workflow in about 15 focused hours across 15 days.
- Implemented responsive class/pricing sections, rental information, enquiry submission, WhatsApp handoff, and SEO basics.

SELECTED PROJECTS

MERLIN - Medical Emergency Routing and Live Intelligence Network

Tech: Python, pandas, NumPy, PuLP, simulation, routing logic, Matplotlib, Seaborn

- Building a healthcare AI simulation for rural/remote emergency dispatch, severity prediction, routing, and resource decision support.
- Includes simulation modules, dispatch logic, safety validation, trained model artifacts, simulation CSV, and six visual result figures.
- Presented responsibly as research-support simulation, not a live clinical system. GitHub: github.com/rajasatyavardhan/merlin-dispatch

AR Glasses AI Assistant / Humber Capstone

Tech: Python, Flutter, Whisper concepts, OpenAI Vision, ChatGPT, Google Vision/Gemini concepts, API design

- Designed a multimodal assistant concept for Brilliant Labs Frame AR glasses using voice input, image understanding, and LLM responses.
- Produced capstone report, presentation, architecture flow, and demo video assets for hands-free contextual assistance.

Predicting Flight Delays at Scale

Tech: PySpark, Random Forest, HDFS, GCP VM, Logstash, Elasticsearch, Kibana, Python

- Built an academic big-data ML pipeline using 60M+ U.S. flight records for delay prediction and operational analytics dashboards.
- Reported academic project metrics: accuracy about 99.27%, F1 about 99.28%, ROC-AUC about 99.79%; not claimed as production validation.

Yelp Sentiment Analysis with LSTM + DistilBERT

Tech: Python, TensorFlow/Keras, PyTorch, Hugging Face, DistilBERT, scikit-learn

- Compared LSTM and DistilBERT models for positive/neutral/negative Yelp review sentiment classification.
- DistilBERT achieved stronger reported performance, approximately 73.4% accuracy and 73.6% F1, with reports and slides available.

Smart Attendance with Face Recognition + Blink Liveness

Tech: Python, OpenCV, DeepFace, MediaPipe, dlib, Kaggle LFW, CSV logging

- Built a computer vision attendance prototype combining face recognition with blink-based liveness detection to reduce photo spoofing risk.
- Used facial landmarks and eye aspect ratio logic for applied biometrics and access-control workflows.

The Dansvilla Studio Website & Growth Automation

Tech: HTML, CSS, JavaScript, Vercel, GitHub, forms, WhatsApp workflow, SEO basics

- Solo-built a live business website and lead workflow for an Ottawa dance studio from real requirements and feedback.
- Live: www.thedansvillastudio.com | GitHub: github.com/rajasatyavardhan/thedansvillastudio

ADDITIONAL AI / IOT PROJECTS

- Synthetic Chest X-ray Augmentation with GANs - PyTorch/DCGAN medical dataset augmentation with SSIM, PSNR, and FID-style evaluation.
- Telugu-English Hate Speech Detection - code-mixed NLP safety project; report complete, code evidence still needs cleanup.
- Visitor Monitoring & Smart Door System - Sheridan IoT project using ESP32, ultrasonic sensors, servo gate control, Blynk dashboard, and threshold alerts.
- Sign Language Recognition using CNN-LSTM - B.Tech capstone project using temporal gesture recognition; public evidence needs name/report cleanup before linking.
- Offline ChatGPT / Local LLM Deployment - Sheridan project using WSL, Ollama, Docker, and OpenWebUI for private local AI experimentation.

EDUCATION

Graduate Certificate - Artificial Intelligence with Machine Learning | Humber Polytechnic, Ontario | 2025

- GPA noted in prior resumes: 85.6%; capstone: AR Glasses AI Assistant

Graduate Certificate - Internet of Things with Machine Learning | Sheridan College, Ontario | 2024

- Focus: ESP32 systems, sensor integration, ML-driven automation

B.Tech - Information Technology | Vellore Institute of Technology, India | 2023

- Capstone/project experience in deep learning and applied systems

CAREER FAIR POSITIONING

- Target roles: AI Developer, Machine Learning Developer, Python Developer, Data/AI Analyst, Software Developer, Applied AI Developer.
- Portfolio theme: Proof over buzzwords - practical systems across healthcare AI, multimodal AI, big data, NLP, computer vision, IoT, and business automation.
- Available for full-time opportunities in Canada; open to Ottawa, hybrid, remote, and relocation-friendly roles.